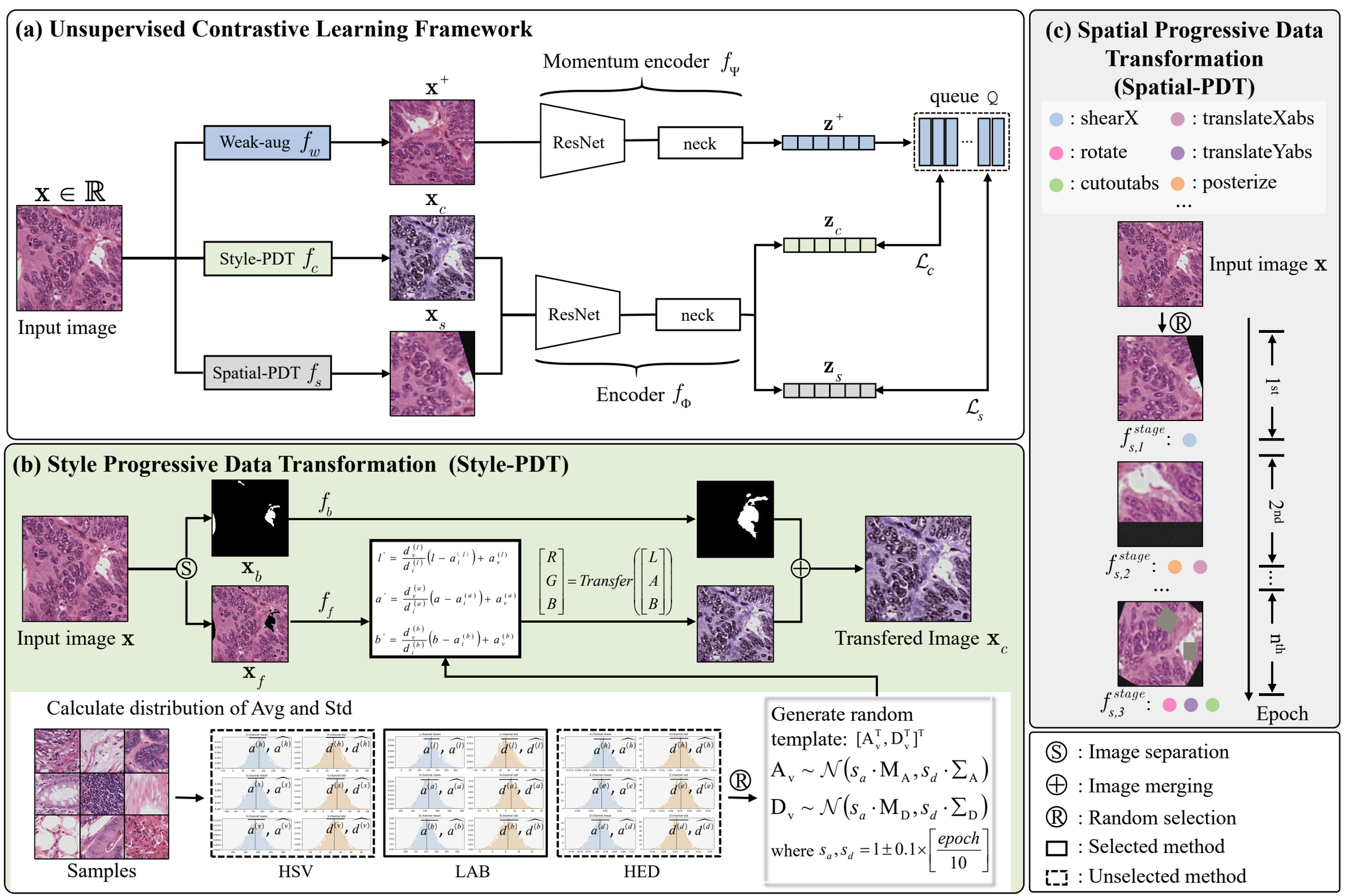
(a) Unsupervised Contrastive Learning Framework
Momentum encoder $f_\Psi$
queue Q
$\mathbf{x} \in \mathbb{R}$
Input image
Weak-aug $f_w$
Style-PDT $f_c$
Spatial-PDT $f_s$
$\mathbf{x}^+$
$\mathbf{x}_c$
$\mathbf{x}_s$
ResNet
neck
$\mathbf{z}^+$
$\mathbf{z}_c$
$\mathbf{z}_s$
$\mathcal{L}_c$
$\mathcal{L}_s$
Encoder $f_\Phi$
(b) Style Progressive Data Transformation (Style-PDT)
Input image $\mathbf{x}$
$\mathbf{x}_b$
$\mathbf{x}_f$
$f_b$
$f_f$
$l' = \frac{d_v^{(l)}}{d_i^{(l)}}(l - a_i^{(l)}) + a_v^{(l)}$
$a' = \frac{d_v^{(a)}}{d_i^{(a)}}(a - a_i^{(a)}) + a_v^{(a)}$
$b' = \frac{d_v^{(b)}}{d_i^{(b)}}(b - a_i^{(b)}) + a_v^{(b)}$
$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = Transfer\left(\begin{bmatrix} L \\ A \\ B \end{bmatrix}\right)$
Transfered Image $\mathbf{x}_c$
Calculate distribution of Avg and Std
Samples
HSV
LAB
HED
Generate random template: $[\mathrm{A}_v^T, \mathrm{D}_v^T]^T$
$\mathrm{A}_v \sim \mathcal{N}(s_a \cdot \mathrm{M}_A, s_d \cdot \Sigma_A)$
$\mathrm{D}_v \sim \mathcal{N}(s_a \cdot \mathrm{M}_D, s_d \cdot \Sigma_D)$
where $s_a, s_d = 1 \pm 0.1 \times \left\lceil \frac{epoch}{10} \right\rceil$
(c) Spatial Progressive Data Transformation (Spatial-PDT)
: shearX
: translateXabs
: rotate
: translateYabs
: cutoutabs
: posterize
...
Input image $\mathbf{x}$
$f_{s,1}^{stage}$ :
$f_{s,2}^{stage}$ :
$f_{s,3}^{stage}$ :
1st
2nd
nth
Epoch
Ⓢ : Image separation
⊕ : Image merging
Ⓡ : Random selection
: Selected method
: Unselected method